

John Mel Renzo Ngo

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EXPERIENCE

Software Engineering Contractor | University of Houston - Clear Lake - Contracted to Boeing | Houston, TX |

June 2024 - May 2025

- Streamlined database verification processes, accelerating data audit completion times by 30% through automated scripts implemented in Python to efficiently flag mismatched items between blueprints and digital documentation.
- Collaborated with cross-functional engineering and IT teams to troubleshoot data discrepancies, ensuring alignment between physical blueprints and digital repositories for seamless compliance during internal audits.
- Managed semester-long data migration project in an Agile environment, overseeing weekly sprint planning and daily standups to integrate over 15,000 blueprint files using Python and SQL, achieving a 42% reduction in manual processing hours while meeting all audit compliance KPIs.

Software Engineering Internship | BJSS | Houston, TX | January 2024 - May 2024

- Built and deployed front-end components for three proprietary applications leveraging HTML, CSS, JavaScript, React, and Node.js, driving a 25% increase in user task completion rate within four months.
- Reduced project delivery timelines by 18% over 5 months by structuring collaborative front-end feature sprints in Agile Scrum, prioritizing high-impact backlog items using React, Node.js, HTML, CSS, and JavaScript.
- Managed source code control for three ongoing projects over a 5-month internship using Git and GitHub, ensuring 100% adherence to branch management workflows and reducing merge conflict frequency by 40%.

PROJECT

Hardware Generation of Numerical Hardware for Scientific Computing | UHCL | August 2024 - May 2025

- Automated code generation pipeline for scientific hardware acceleration projects, streamlining deployment to FPGA platforms and shortening development cycles by 20%, employing Chisel, Vivado, and integrated version control.
- Increased verification throughput by 35% by deploying ModelSim and UVM-based automated simulation frameworks, reducing test cycle times from 2 days to under 14 hours for numerical hardware utilized in scientific computing workflows.
- Engineered parameterized hardware generators using Chisel and Scala, enabling on-demand RTL customization across 12+ scientific computing projects and boosting design reuse by 40%, while supporting rapid iterations, automated regression testing, and seamless integration with Vivado toolchains.

Honeycomb - Inventory Manager | Honeybee Bakery | June 2024 - May 2025

- Engineered Honeycomb - Inventory Manager to automate stock monitoring, consolidate vendor data, and enhance order fulfillment workflows for increased operational agility.
- Integrated predictive analytics using Python and SQL to forecast inventory needs, achieving a 25% reduction in stockouts and a 15% decrease in holding costs over a 12-month period while supporting seamless multi-vendor coordination.
- Instituted an intuitive dashboard enabling real-time inventory tracking, centralized reporting, and actionable insights to empower data-driven decision-making across supply chain teams.

Rocket Launch | JSC NASA Hackathon | github.com/TruongLai28/JSC_Hack | November 2024 - November 2024

- Processed and centralized over 300 NASA-affiliated opportunities using API-driven data ingestion and SQL data models, streamlining access and lowering user search effort by 35% across the pilot deployment period.
- Increased page load efficiency by 30% through the deployment of a Next.js and React-based platform with Tailwind CSS, optimizing load balancing and resource management for 5,000+ concurrent users monthly.
- Engineered a back-end system using FastAPI, achieving a 45% reduction in search result latency by integrating Playwright for high-frequency web data extraction and leveraging ChromaDB for efficient data querying.

EDUCATION

B.S., Computer Engineering | University of Houston-Clear Lake | Houston, TX | 2025 | 3.74

CERTIFICATIONS

AWS Certified AI Practitioner (In Progress) | AWS

SKILLS

Hard Skills: Web Development, Embedded Systems, Data Analysis & Visualization, Linux Systems, Agile Methodology, AI, Machine Learning, Cloud
Technical Skills: Python, C/C++, JavaScript, Next.js, Node.js, API, Power BI, Tableau, MATLAB, SQL, Git, MS Office Suite, RAG, LangChain, AWS
Soft Skills: Problem Solver, Analytical Thinker, Attention To Detail, Adaptable, Fast Learner, Leadership, Team Collaboration & Cross-Disciplinary Communication